BIOL 343
Applied Bioinformatics

Post-alignment QC

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Alignment is the most important step in RNA-seq analysis

Counting (also important) and DEG ID relies on high-confidence mapping

- Recall the goal of our RNA-seq experiments…
  - Treatment vs Control
  - Mutant vs Wild type
  - ***Identify differentially expressed genes (DEGs)***
- DEGs will be identified using statistical tests comparing ***expression values*** of transcripts/genes
- Expression values will be calculated based on the number of reads that ***align/map*** to a given genomic locus
- ***Filtering*** out spurious or redundant alignments is an important QC step after mapping

Browsing RNA-seq alignments

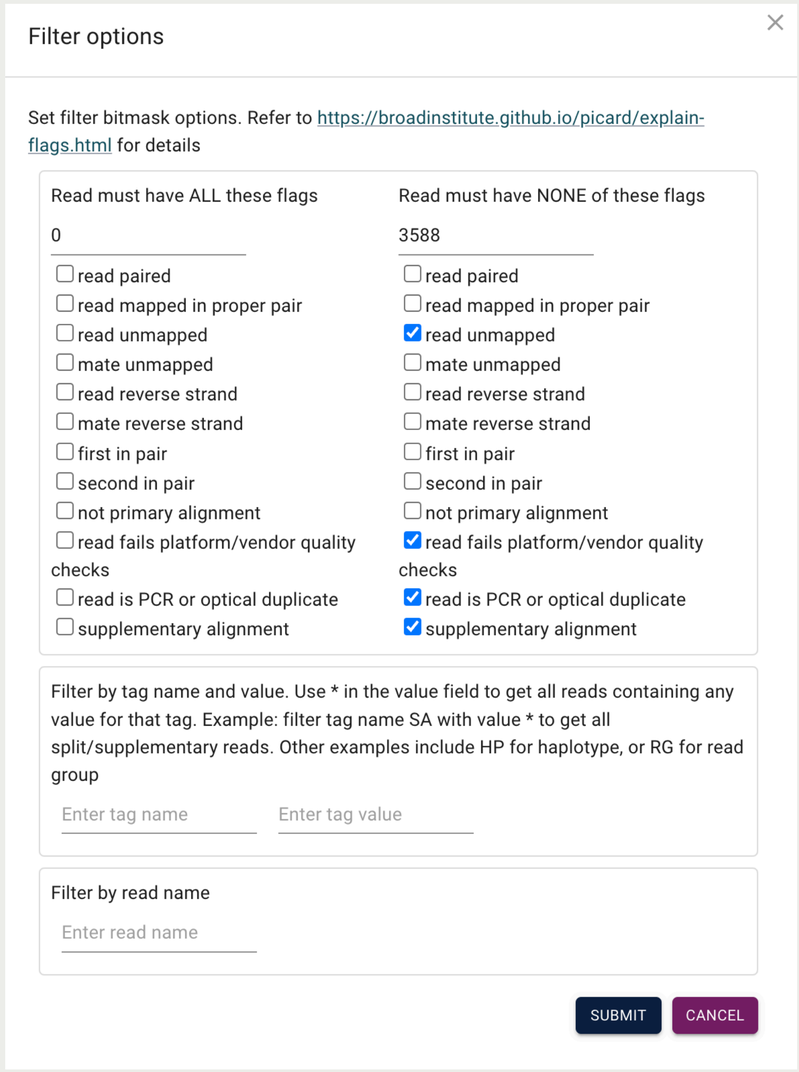
JBrowse2

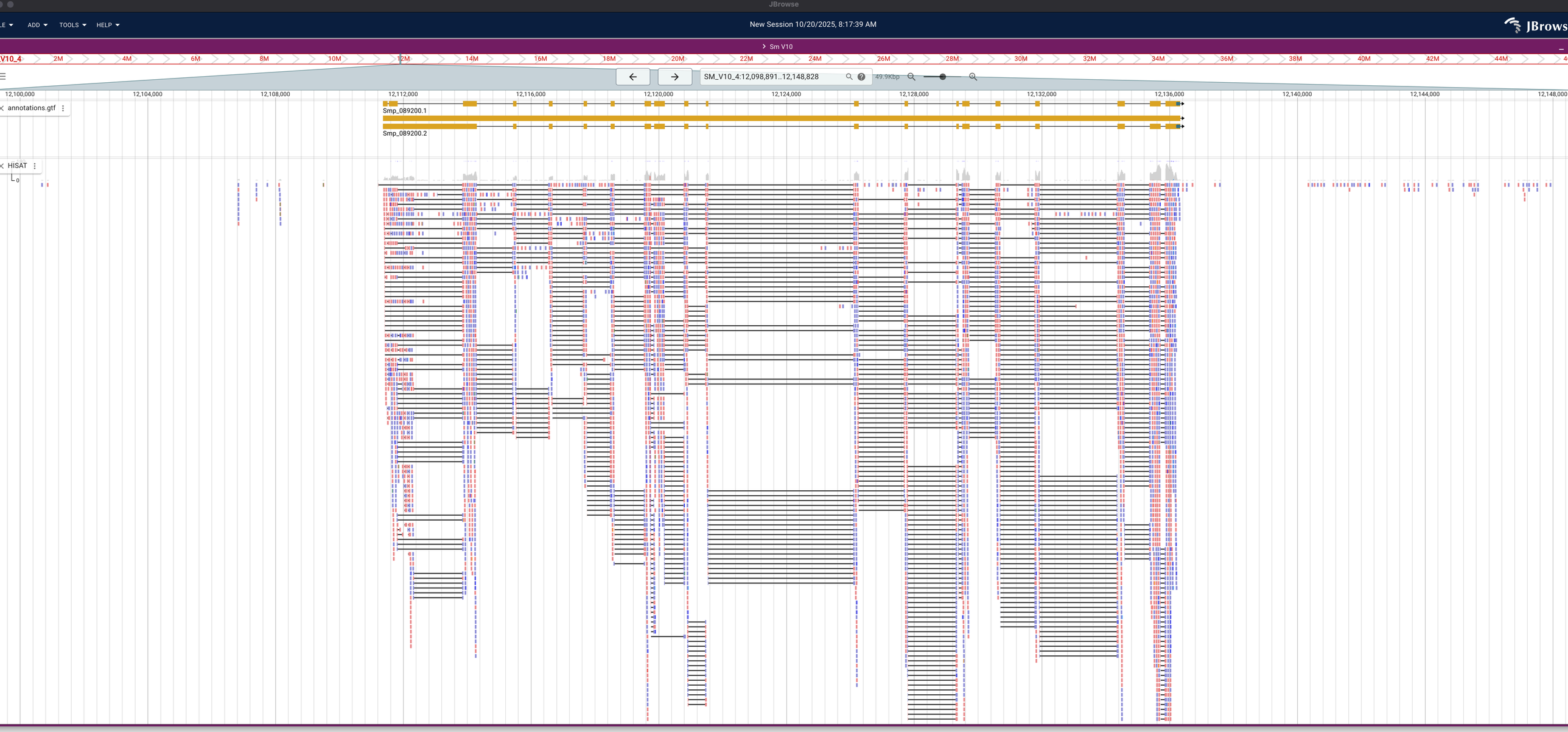
1. Open JBrowse2
2. Download the genome files (.fa & .gtf)
3. Download the STAR alignment (Aligned.sorted.bam) and indexes
4. Load the genome (.fa)
5. Add tracks:
    1. GTF
    2. Alignments

Browsing RNA-seq alignments

JBrowse2

1. Navigate to NC_141034.1:11961474..11964958
2. Three dots on RNA-seq track…
    1. Pileup settings > Filter by…
    2. Pileup settings > Color by > Strand
    3. SNPCoverage settings > Draw arcs (deselect)




JBrowse
ADD
TOOLS
HELP
New Session 10/20/2025, 8:17:39 AM
JBrows
Sm V10
SM_V10_4:12,098,891..12,148,828
49.9Kbp
annotations.gtf
Smp_089200.1
Smp_089200.2
HISAT